COMPUTER ENGINEERING

QUEZON CITY

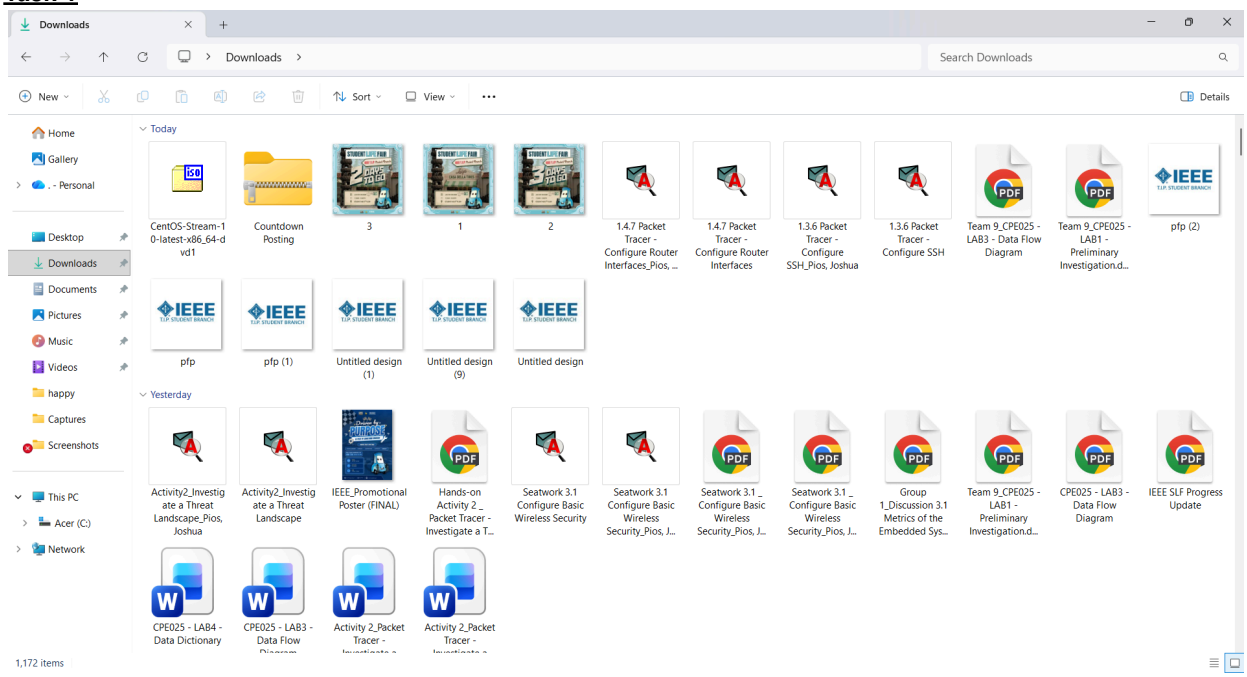
COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	July 16, 2026	**DATE SUBMITTED:**	July 16, 2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 2.1 SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

What procedures did you perform?

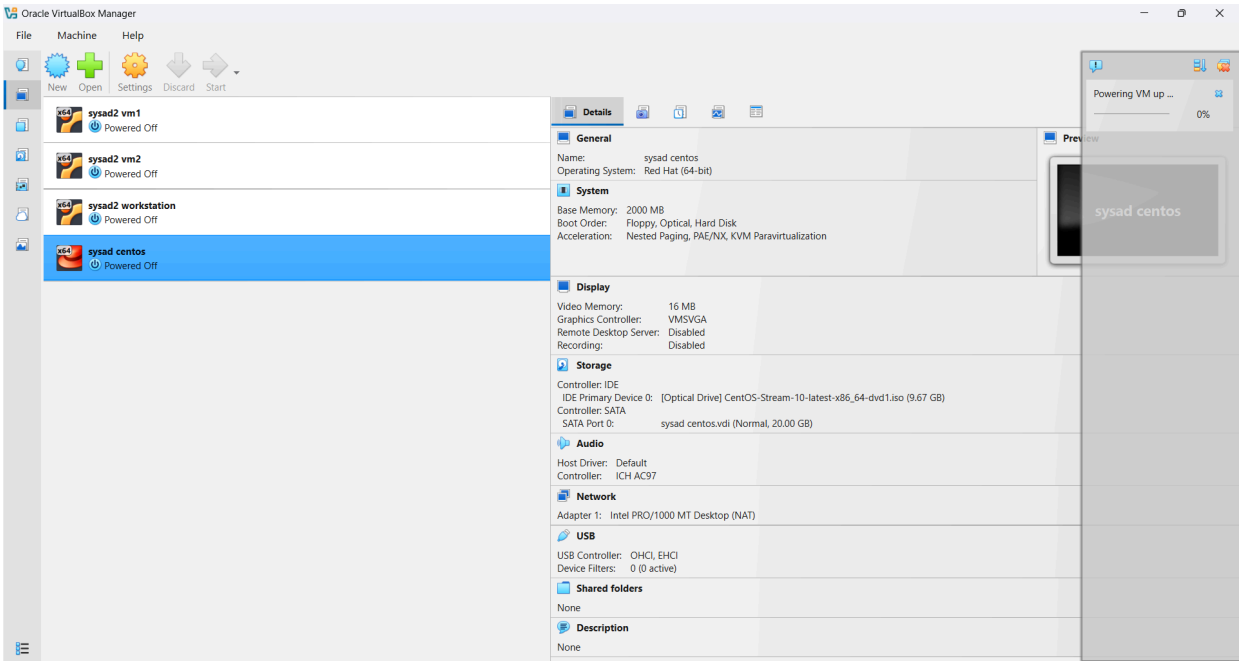
Task 1

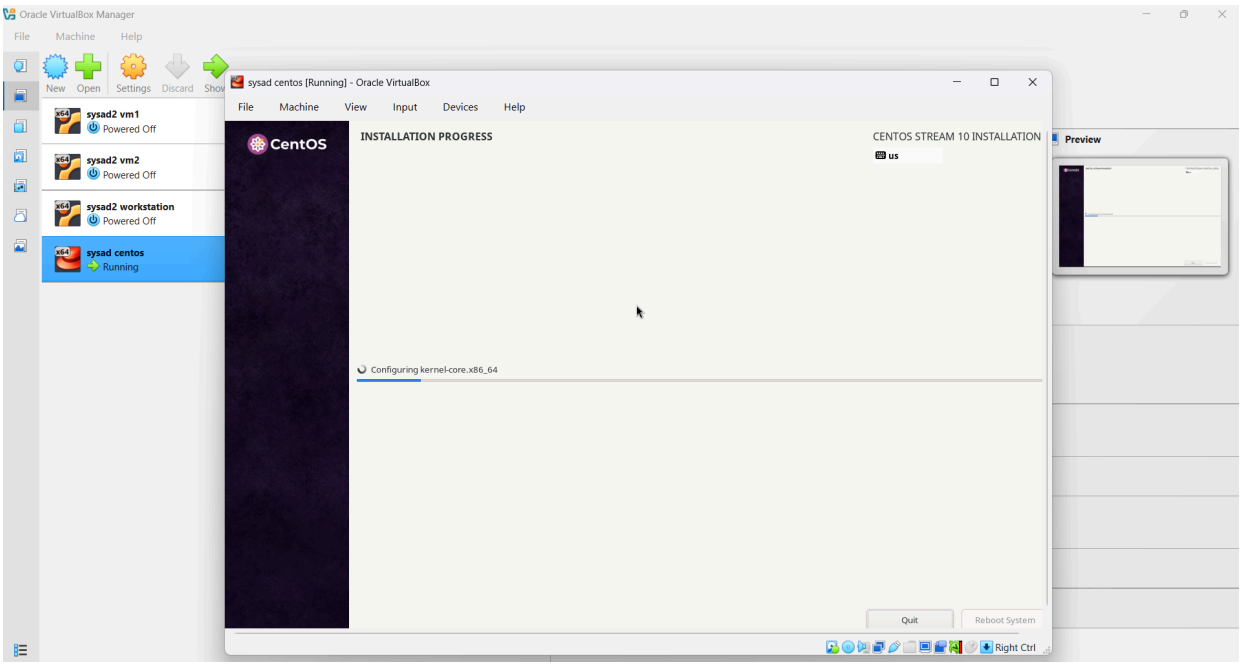


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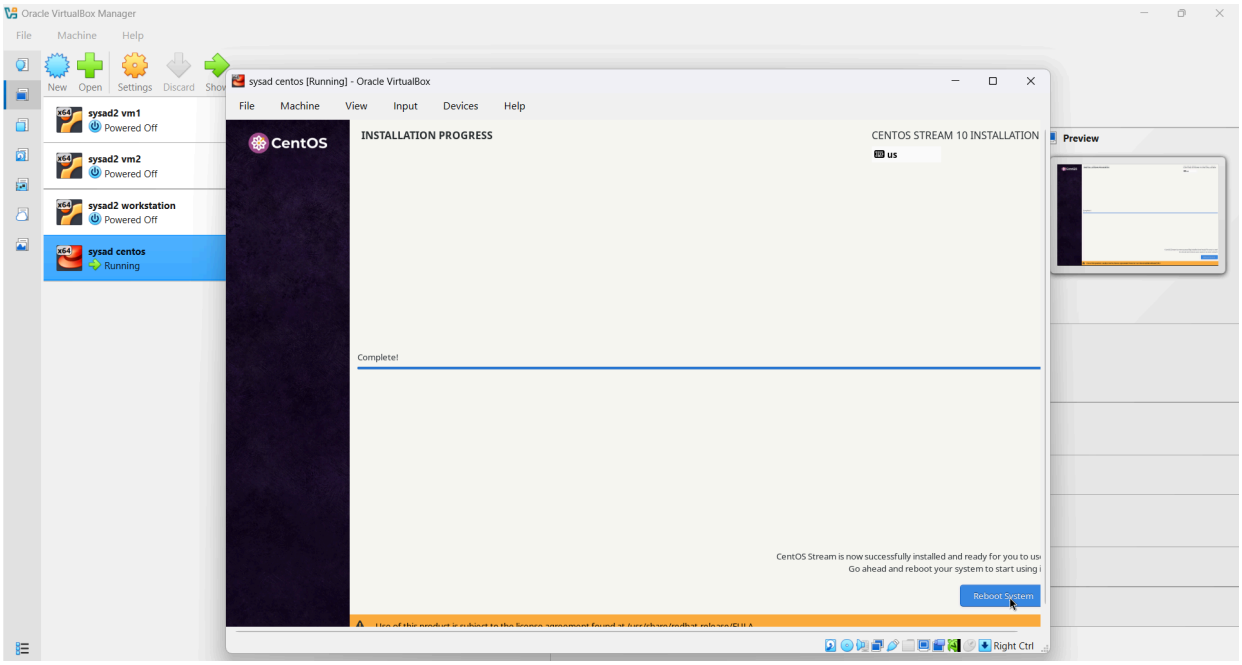


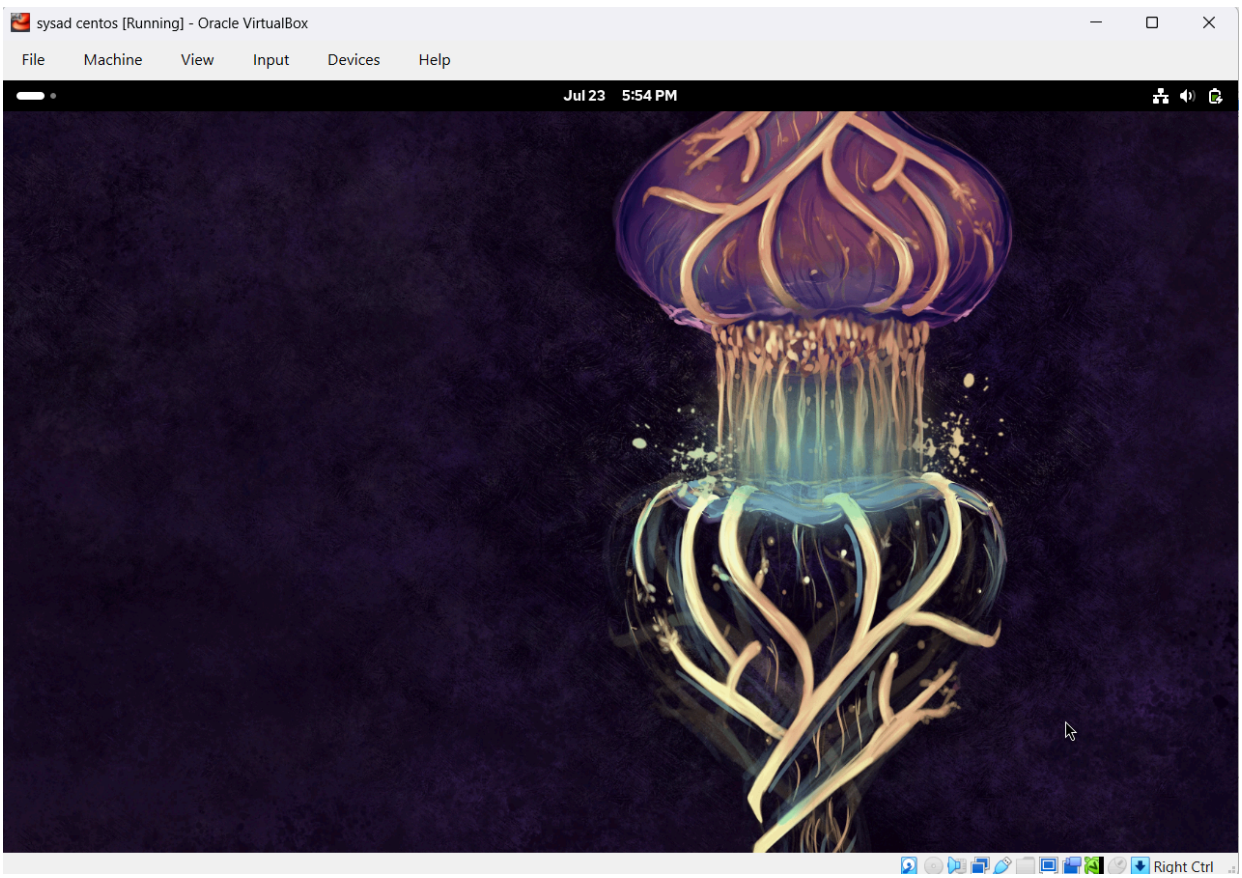


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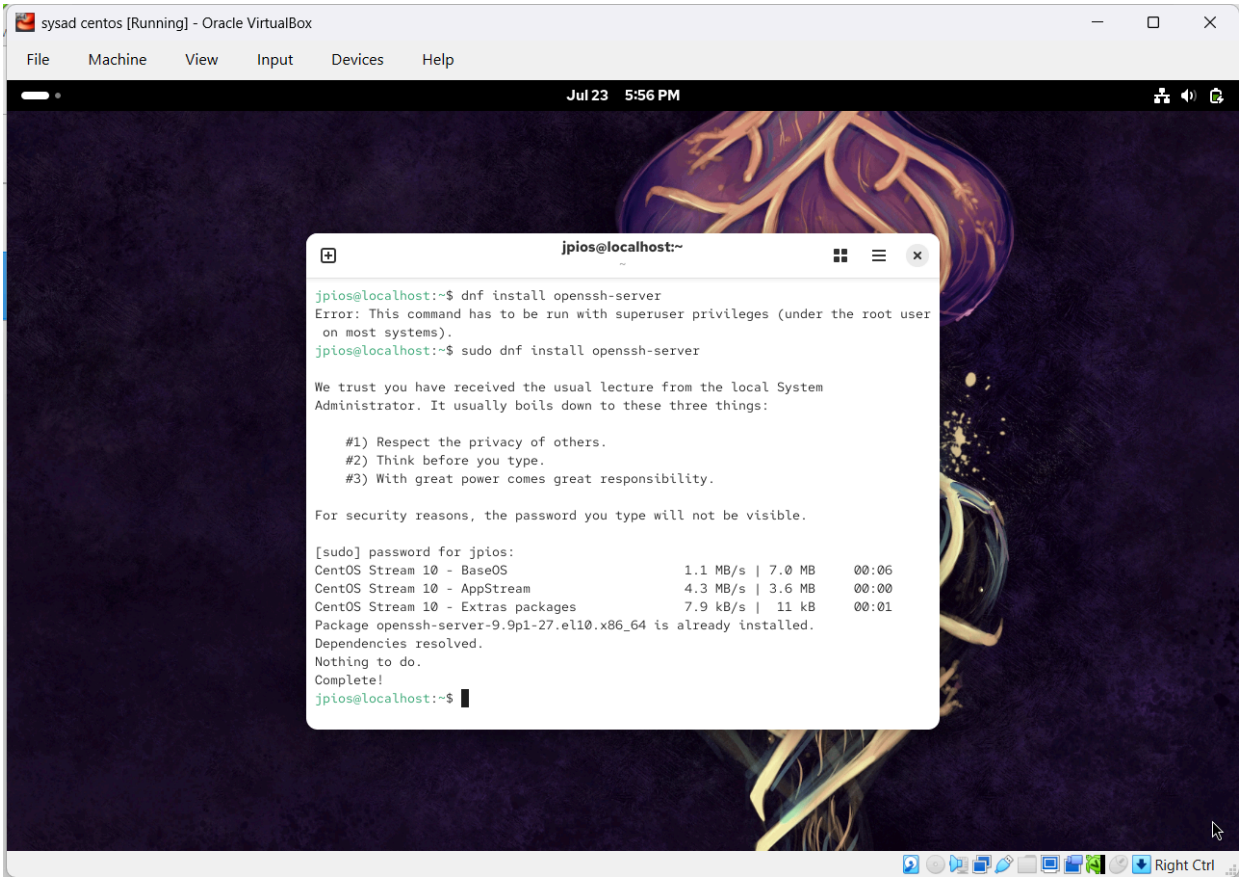
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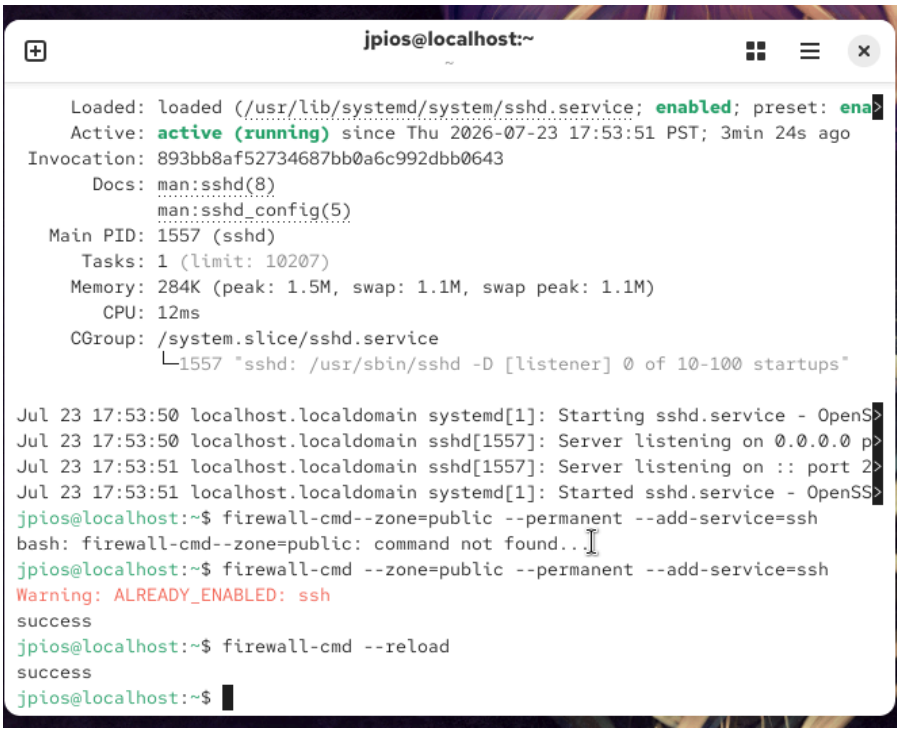




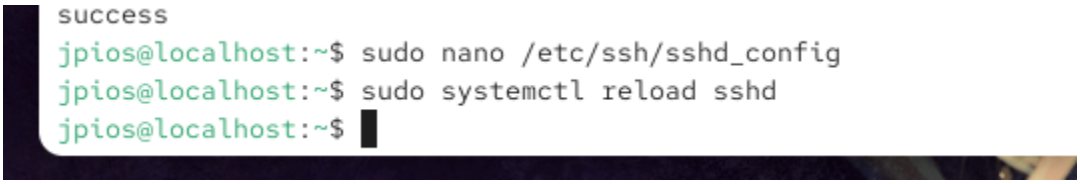
Task 2


sysad centos [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Jul 23 5:56 PM
jpios@localhost:~
jpios@localhost:~$ dnf install openssh-server
Error: This command has to be run with superuser privileges (under the root user on most systems).
jpios@localhost:~$ sudo dnf install openssh-server
We trust you have received the usual lecture from the local System Administrator. It usually boils down to these three things:
#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.
For security reasons, the password you type will not be visible.
[sudo] password for jpios:
CentOS Stream 10 - BaseOS 1.1 MB/s | 7.0 MB 00:06
CentOS Stream 10 - AppStream 4.3 MB/s | 3.6 MB 00:00
CentOS Stream 10 - Extras packages 7.9 kB/s | 11 kB 00:01
Package openssh-server-9.9p1-27.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
jpios@localhost:~$
Right Ctrl

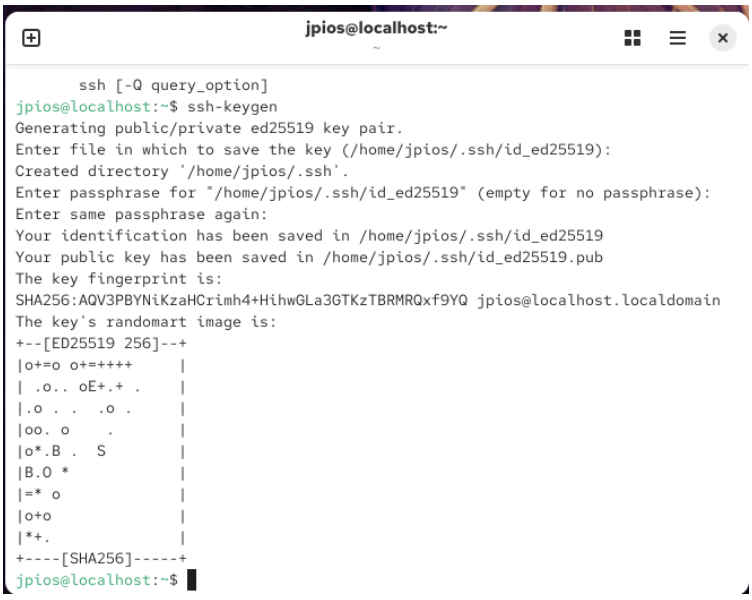

jpios@localhost:~
Dependencies resolved.
Nothing to do.
Complete!
jpios@localhost:~$ systemctl start sshd
jpios@localhost:~$ systemctl enable sshd
jpios@localhost:~$ systemctl status sshd
● sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena
Active: active (running) since Thu 2026-07-23 17:53:51 PST; 3min 24s ago
Invocation: 893bb8af52734687bb0a6c992dbb0643
Docs: man:sshd(8)
man:sshd_config(5)
Main PID: 1557 (sshd)
Tasks: 1 (limit: 10207)
Memory: 284K (peak: 1.5M, swap: 1.1M, swap peak: 1.1M)
CPU: 12ms
CGroup: /system.slice/sshd.service
└1557 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
Jul 23 17:53:50 localhost.localdomain systemd[1]: Starting sshd.service - OpenS
Jul 23 17:53:50 localhost.localdomain sshd[1557]: Server listening on 0.0.0.0 p
Jul 23 17:53:51 localhost.localdomain sshd[1557]: Server listening on :: port 2
Jul 23 17:53:51 localhost.localdomain systemd[1]: Started sshd.service - OpenSS
jpios@localhost:~$


jpios@localhost:~
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena
Active: active (running) since Thu 2026-07-23 17:53:51 PST; 3min 24s ago
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Jul 23 17:53:51 localhost.localdomain sshd[1557]: Server listening on :: port 2
Jul 23 17:53:51 localhost.localdomain systemd[1]: Started sshd.service - OpenSS
jpios@localhost:~$ firewall-cmd--zone=public --permanent --add-service=ssh
bash: firewall-cmd--zone=public: command not found...
jpios@localhost:~$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
jpios@localhost:~$ firewall-cmd --reload
success
jpios@localhost:~$


jpios@localhost:~ – sudo nano /etc/ssh/sshd_config
GNU nano 8.1 /etc/ssh/sshd_config
$OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $
This is the sshd server system-wide configuration file. See
sshd_config(5) for more information.
This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr
The strategy used for options in the default sshd_config shipped with
OpenSSH is to specify options with their default value where
possible, but leave them commented. Uncommented options override the
default value.
To modify the system-wide sshd configuration, create a *.conf file under
/etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf
If you want to change the port on a SELinux system, you have to tell
SELinux about this change.
semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


success
jpios@localhost:~$ sudo nano /etc/ssh/sshd_config
jpios@localhost:~$ sudo systemctl reload sshd
jpios@localhost:~$

Task 3


jpios@localhost:~
ssh [-Q query_option]
jpios@localhost:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/jpios/.ssh/id_ed25519):
Created directory '/home/jpios/.ssh'.
Enter passphrase for "/home/jpios/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/jpios/.ssh/id_ed25519
Your public key has been saved in /home/jpios/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:AQV3PBYNiKzaHCrimh4+HihwGLa3GTKzTBRMRQxf9YQ jpios@localhost.localdomain
The key's randomart image is:
+--[ED25519 256]--+
|o+=o o+=++++ |
| .o.. oE+.+ . |
|.o . . .o . |
|oo. o . |
|o*.B . S |
|B.O * |
|=* o |
|o+o |
|*+. |
+----[SHA256]-----+
jpios@localhost:~$

```
jpios@localhost:~$ ssh-copy-id jpios@10.0.2.15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is SHA256:sAiGu+fsYqdYpEBhJb4yHG0412ZWf4v4kkF+dZb4UIo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
jpios@10.0.2.15's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'jpios@10.0.2.15'"
and check to make sure that only the key(s) you wanted were added.

jpios@localhost:~$
```

```
jpios@localhost:~$ ls ~/.ssh
authorized_keys  id_ed25519  id_ed25519.pub  known_hosts  known_hosts.old
jpios@localhost:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

The best Linux distribution depends on the intended use, ease of management, software compatibility, and available support. Debian is a good choice for stability and community support, while Red Hat-based distributions are better suited for enterprise environments with professional support and security features.

2. What are the main differences between Debian and Red Hat Linux distributions?

The main difference is that Debian uses the APT package manager with DEB packages, while Red Hat uses DNF/YUM with RPM packages. Red Hat distributions are commonly used in enterprise environments, whereas Debian is known for its stability, flexibility, and large community support.

Conclusion/Learnings:

This activity provided hands-on experience in installing a CentOS operating system and configuring an SSH server for secure remote access. It also demonstrated the importance of properly managing system services, firewall settings, and SSH authentication to establish secure communication between machines. Overall, the activity strengthened my understanding of Linux server administration and the essential role of SSH in remote system management.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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